

to two downstream tasks: protein family classification and protein function prediction.

Protein Family Classification. We assess ProtoMech on a collection of binary classification problems corresponding to protein family membership. We construct these tasks using Swiss-Prot sequences clustered at 30% sequence identity and annotated with InterPro family labels (Paysan-Lafosse et al., 2022). For each family, we train a logistic regression probe on $\mathbf{y}^L$, which serves as the reference computation that the ProtoMech circuit must reproduce.

Function Prediction. We next evaluate ProtoMech’s ability to model protein fitness landscapes. We select 12 Deep Mutational Scanning (DMS) assays from ProteinGym (Notin et al., 2023a), spanning a diverse set of biological functions (see Table 2 for the full list of assays). For each DMS assay, we adopt the supervised cross-validation protocols provided by ProteinGym, which define two evaluation regimes: (i) training and testing exclusively on single mutants, and (ii) training and testing on both single and higher-order mutants. To establish the performance of the original model, we train a convolutional neural network (CNN) probe on the final-layer MLP output $\mathbf{y}^L$, following the training procedure described in (Notin et al., 2023b).

Circuit Discovery Algorithm. Similar to (Nainani et al., 2025; Dunefsky et al., 2024), to recover the minimal subset of latent variables required for each task, we employ an iterative greedy search procedure based on gradient-based attribution. For each latent unit, we compute an attribution score by measuring its contribution to the probe’s output on a held-out validation set. Latents are then ranked by attribution magnitude and incrementally added to the candidate circuit in small batches. This procedure continues until the resulting circuit recovers at least 70% of the original ESM2 performance, or until it reaches the performance of the full replacement model when all latents are included. We use the F1 score as the evaluation metric for protein family classification and the Spearman rank correlation for function prediction. Additional details of the circuit discovery procedure are provided in Appendices D.1 and D.2.

2.3. Steering

To further validate the computational pathways discovered by ProtoMech, we intervene on the function prediction circuits to steer ESM2 toward high-fitness sequences (Fig. 2b). We focused our analysis on seven of the original 12 DMS assays, selected based on the availability of sufficient mutant data. To independently evaluate the fitness of the generated variants, we trained a CNN evaluation model on 90% of the training data, utilizing the same architecture and training procedure as our circuit discovery CNN probes.

Our steering strategy involves activation clamping within

the CLT replacement model, similar to (Templeton et al., 2024). During the forward pass of a wildtype sequence, we select a subset of nodes within the target function circuit and fix their activations to a high activation. We choose this high activation by identifying the maximum activation magnitude observed for that node across the entire sequence, and multiply it by a scalar multiplier. We then utilize Equation (2), setting $\ell = L$, to obtain $\hat{\mathbf{y}}^L$ and then decode the resulting representation to ESM2’s logits. We select mutations corresponding to the maximum probability values. To ensure the CNN evaluation model remains a valid proxy for experimental fitness, we constrained the generation to variants within a radius of five mutations from the wildtype. We note that steering via Equation (2) and setting $\ell = L$ is one of many possible ways, and we provide a comprehensive overview of all the steering methods tested in Appendix D.3. We report our results steering ProtoMech and PLT circuits following an ablation study conducted in Appendix E.

2.4. Visualizing Circuits

To qualitatively analyze the computational pathways discovered, we construct sparse graph representations using the ProtoMech visualizer (Fig. 2c). For a given input sequence, we isolate the graph by selecting the top-five highest activating nodes per layer. When analyzing the mutational effects of sequences in function prediction circuits, we additionally include the top-five nodes exhibiting the largest activation shift between the wildtype and variant.

From here, we aim to identify possible biological overlap learned by each of the nodes. We identify the specific amino acids that are being activated in the input sequence and cross-reference them with the top-10 maximally activating sequences from Swiss-Prot to detect conserved motifs or family-specific patterns (Fig. 1b). We further project these activated amino acids onto the protein structure to visualize their spatial clustering and infer their biological relevance. Finally, to trace the computational path learned by ESM2, we connect our graph with edges following (Ameisen et al., 2025). Briefly, we compute the edge weight connecting a source node to a target node, where the target node must reside at a subsequent layer. This weight is calculated as the product of the source node’s activation and the gradient of the target’s pre-activation with respect to the source node (see Appendix F for more details).

3. Results

We evaluate the performance of ProtoMech on three distinct tasks: (i) circuit discovery, (ii) steering, and (iii) uncovering overlapping biological motifs in circuits. For additional experimental results, including tests of other replacement models described in Appendix C and our steering ablation study, we refer to Appendix E.

3.1. ProtoMech Compresses ESM2’s Computation

Protein Family Circuits. We first evaluated the accuracy of ProtoMech on protein family classification tasks (Fig. 1c). Using the full set of latents, ProtoMech achieves an average F1 score of 0.82 ± 0.19 , recovering approximately 89% of the original model’s performance (0.92 ± 0.10). This significantly outperforms the PLT baseline, which achieved an average F1 of 0.50 ± 0.34 , suggesting that the underlying representation learned by ProtoMech provides a substantially more accurate approximation of ESM2’s computation.

This performance advantage extends to circuit discovery. The circuits identified by ProtoMech achieve an average F1 of 0.73 ± 0.19 , (79% recovery compared to the original model), compared to 0.49 ± 0.33 from the PLT. Notably, ProtoMech achieves this while using only 150 ± 204 latents on average (roughly 0.8% of the total possible latent space), demonstrating its ability to compress biologically relevant information into a sparse set of latents.

Function Prediction Circuits. We observe similar trends when applying ProtoMech to fitness regression tasks (Fig. 1d). On the full set of latents, ProtoMech achieves an average Spearman correlation of 0.41 ± 0.19 , recovering approximately 82% of the original model’s performance (0.50 ± 0.22). ProtoMech again outperforms the PLT baseline, which achieved an average Spearman of 0.38 ± 0.18 .

When identifying circuits, ProtoMech retains its advantage, achieving an average Spearman of 0.38 ± 0.18 (74% recovery) compared to 0.35 ± 0.19 for the PLT. ProtoMech again achieves this recovery while using only a tiny fraction of the possible latent space (167 ± 221 latents, making up 0.9% of the total latent space).

3.2. ProtoMech Designs High-fitness Sequences

We next evaluated whether the computational pathways identified by ProtoMech could be used to steer the model toward high-fitness variants. To benchmark our approach, we compared it with steering using PLT circuits, selecting random mutations, and Contrastive Activation Addition (CAA) (Huang et al., 2025), a popular method for steering dense pLMs without a sparse latent representation. Briefly, CAA identifies a global “concept vector” in ESM2’s hidden dimension by computing the difference between the mean activations (y^L) of high-fitness and low-fitness sequences; this vector is then injected into the residual stream during generation to bias the model’s output (see Appendix D.4).

Table 1 presents our steering results on the seven selected DMS assays. Interestingly, we find that steering via circuits (both ProtoMech and PLT) consistently yields sequences with higher fitness than CAA, and that circuit-based methods outperform CAA and random in all metrics. We attribute the limitations of CAA to data scarcity and the mutation-

locality of DMS assays. First, CAA relies on averaging activations to estimate a robust direction of improvement. However, DMS datasets are often relatively small (containing as few as 1,000 high-fitness sequences in some splits), which may be insufficient to estimate a concept vector. Secondly, these sequences are extremely localized around the wildtype, typically only one to two mutations away. Consequently, the concept vector is likely overfitting to this localized region and is unable to extrapolate effectively to high-fitness regions of the fitness landscape. In contrast, circuit-based methods leverage a sparse, biologically disentangled latent space. By identifying and clamping only the specific nodes that govern protein function, steering with circuits minimizes interference from irrelevant features, allowing for higher fidelity in targeted generation. When comparing both ProtoMech and PLT directly, we find that ProtoMech outperforms PLT in 71% of cases, which includes generating the single highest-fitness variant, as well as the highest top 10% and top 20% fitness variants. This suggests that ProtoMech captures functional dependencies more effectively than PLT and can generate a diverse pool of highly functional variants.

3.3. ProtoMech Reveals Known Biological Motifs

Given the performance of ProtoMech, a natural question arises: as we trace the computational pathways of ESM2, do we uncover known *biological* motifs? To this end, we perform a rigorous qualitative analysis on our circuits, focusing on three representative case studies: protein kinase and NADP+ binding domains (from protein family classification) and the GB1 fitness landscape (from function prediction). Our analysis reveals that ProtoMech successfully disentangles ESM2 into interpretable computational pathways that overlap with known biological motifs.²

Protein Kinase Domain. Fig. 3a depicts a circuit discovered governing the protein kinase domain (InterPro ID IPR000719), which catalyzes the chemical reactions that regulate cell signaling and behavior (Taylor & Kornev, 2011). We observe that earlier layers tend to recognize specific amino acids that make up more complex motifs in later layers. Specifically, layer 1 at latent 1582 (L1/1582) activates on arginine (R) amino acids, which serve as essential building blocks for catalytic activity (Kornev et al., 2006). This feeds directly into layer 3 (L3/1918), which identifies the HRD motif—a highly conserved catalytic loop responsible for binding to substrate peptides for phosphorylation, the key mechanism for kinase signaling (Modi & Dunbrack, 2019). Subsequently, layer 5 is split into different motifs: L5/1774 identifies the ATP-binding site, a pocket that fa-

²Figs. 3 and 4 display a subset of findings selected for annotation availability and representative cross-layer coverage. These circuits aim to provide an overview of the mechanisms captured by ProtoMech and are by no means exhaustive.

Table 1. Steering results using ProtoMech, PLT, CAA, and selecting random mutations on seven DMS assays. All variants were constrained to a maximum of five mutations away from the wildtype.

Method	DMS	Mean score $\uparrow$	Max score $\uparrow$	Top 10% score $\uparrow$	Top 20% score $\uparrow$
ProtoMech	SPG1_STRSG_Olson_2014	1.67 $\pm$ 0.67	3.24	3.00 $\pm$ 0.21	2.70 $\pm$ 0.34
	HIS7_YEAST_Pokusaeva_2019	1.28 $\pm$ 0.09	1.42	1.41 $\pm$ 0.01	1.40 $\pm$ 0.02
	GRB2_HUMAN_Faure_2021	-0.15 $\pm$ 0.07	0.05	-0.01 $\pm$ 0.04	-0.05 $\pm$ 0.04
	GFP_AEQVL_Sarkisyan_2016	4.17 $\pm$ 0.23	4.63	4.56 $\pm$ 0.05	4.50 $\pm$ 0.08
	CAPSD_AAV2S_Sinai_2021	1.68 $\pm$ 1.08	4.54	3.77 $\pm$ 0.43	3.36 $\pm$ 0.53
	RASK_HUMAN_Weng_2022_abundance	-0.12 $\pm$ 0.06	0.06	0.02 $\pm$ 0.03	-0.02 $\pm$ 0.05
	A4_HUMAN_Seuma_2022	0.12 $\pm$ 0.33	1.07	0.76 $\pm$ 0.18	0.63 $\pm$ 0.19
PLT	SPG1_STRSG_Olson_2014	1.97 $\pm$ 0.48	3.18	2.88 $\pm$ 0.20	2.67 $\pm$ 0.25
	HIS7_YEAST_Pokusaeva_2019	1.27 $\pm$ 0.06	1.41	1.38 $\pm$ 0.02	1.36 $\pm$ 0.03
	GRB2_HUMAN_Faure_2021	-0.18 $\pm$ 0.14	0.15	0.12 $\pm$ 0.05	0.03 $\pm$ 0.09
	GFP_AEQVL_Sarkisyan_2016	4.40 $\pm$ 0.22	5.15	4.92 $\pm$ 0.19	4.76 $\pm$ 0.21
	CAPSD_AAV2S_Sinai_2021	0.81 $\pm$ 0.53	2.05	1.87 $\pm$ 0.11	1.67 $\pm$ 0.22
	RASK_HUMAN_Weng_2022_abundance	-0.19 $\pm$ 0.08	0.06	-0.02 $\pm$ 0.06	-0.07 $\pm$ 0.07
	A4_HUMAN_Seuma_2022	-0.05 $\pm$ 0.38	1.03	0.67 $\pm$ 0.24	0.52 $\pm$ 0.24
CAA	SPG1_STRSG_Olson_2014	0.70 $\pm$ 0.00	0.70	0.70 $\pm$ 0.00	0.70 $\pm$ 0.00
	HIS7_YEAST_Pokusaeva_2019	0.52 $\pm$ 0.14	0.77	0.77 $\pm$ 0.00	0.71 $\pm$ 0.06
	GRB2_HUMAN_Faure_2021	-0.40 $\pm$ 0.11	-0.20	-0.21 $\pm$ 0.02	-0.23 $\pm$ 0.03
	GFP_AEQVL_Sarkisyan_2016	2.93 $\pm$ 0.23	3.16	3.16 $\pm$ 0.00	3.16 $\pm$ 0.00
	CAPSD_AAV2S_Sinai_2021	-0.26 $\pm$ 0.55	0.45	0.45 $\pm$ 0.00	0.45 $\pm$ 0.00
	RASK_HUMAN_Weng_2022_abundance	-0.35 $\pm$ 0.07	-0.28	-0.28 $\pm$ 0.00	-0.29 $\pm$ 0.01
	A4_HUMAN_Seuma_2022	-1.90 $\pm$ 0.03	-1.87	-1.87 $\pm$ 0.00	-1.87 $\pm$ 0.00
Random	SPG1_STRSG_Olson_2014	-2.76 $\pm$ 0.93	-1.25	-1.37 $\pm$ 0.09	-1.56 $\pm$ 0.22
	HIS7_YEAST_Pokusaeva_2019	0.56 $\pm$ 0.25	1.08	0.93 $\pm$ 0.07	0.88 $\pm$ 0.07
	GRB2_HUMAN_Faure_2021	-0.84 $\pm$ 0.26	-0.37	-0.45 $\pm$ 0.06	-0.50 $\pm$ 0.07
	GFP_AEQVL_Sarkisyan_2016	2.74 $\pm$ 0.51	3.54	3.46 $\pm$ 0.05	3.35 $\pm$ 0.12
	CAPSD_AAV2S_Sinai_2021	-1.04 $\pm$ 0.61	0.50	0.09 $\pm$ 0.39	-0.26 $\pm$ 0.45
	RASK_HUMAN_Weng_2022_abundance	-0.64 $\pm$ 0.26	-0.04	-0.21 $\pm$ 0.09	-0.27 $\pm$ 0.09
	A4_HUMAN_Seuma_2022	-1.57 $\pm$ 0.61	-0.87	-0.94 $\pm$ 0.05	-0.99 $\pm$ 0.07

cilitates the docking of ATP for phosphorylation (Taylor & Kornev, 2011), while L5/826 detects the Glycine-rich loop (G-loop), which anchors ATP during catalysis (Bossemeyer, 1994). Consistent with findings in (Adams et al., 2025), we observe that the final layer also tends to recognize specific amino acids. For instance, layer 6 (L6/886) activates strongly on serine (S) amino acids. One possible explanation is that, because the representative input sequence used to model these activations belongs to the serine/threonine-protein kinase family, ESM2 aggregates structural context learned in earlier layers to pinpoint the specific serine residue required for phosphorylation.

NADP+ Binding Domain. Fig. 3b depicts a circuit governing the NADP+ binding domain superfamily (InterPro ID IPR036291). In this particular example, we focus on the protein Metalloredutase STEAP3, which binds to Nicotinamide Adenine Dinucleotide Phosphate (NADP+) and Flavin Adenine Dinucleotide (FAD) in order to facilitate the electron transfer of iron for the cell (Ohgami et al., 2005). We similarly find that earlier layers tend to recognize specific amino acids. For instance, L1/967 activates on phenylalanine (F) residues, which are present in the NADP+ binding sites. This latent variable feeds into layer 4 (L4/53),

which identifies the Rossmann fold, a structural motif composed of alternating beta strands and alpha helices that facilitate NADP+ binding (Hanukoglu, 2015). From here, this information is fed into layer 5 (L5/1338), which narrows this context into specific NADP+ binding pockets within the Rossmann fold. Separately, layer 5 (L5/1109) also detects NADP+ and FAD-binding sites in the sequence. Finally, in the last layer of ESM2, we observe repeated concepts from earlier layers. For instance, layer 6 (L6/2198) reactivates F amino acids as in layer 1. This recurrence mirrors mechanistic interpretability findings in natural language (Ameisen et al., 2025), suggesting that ESM2 reasserts specific residue details in its predictions.

Mutation Sensitivity in GB1. We evaluated the mutational sensitivity of our discovered circuits toward high- and low-fitness variants, focusing our analysis on the IgG-binding domain of Protein G (GB1). GB1 is a streptococcal surface protein that functions by binding to the Fc fragment of immunoglobulin G (IgG) antibodies (Sloan & Hellinga, 1999). When inputting the wildtype sequence into ProtoMech (Fig. 4a), we found that layer 1 (L1/249) activates on tryptophan (W) at position 43 (W43). W43 is a critical aromatic residue buried in GB1’s hydrophobic core, which